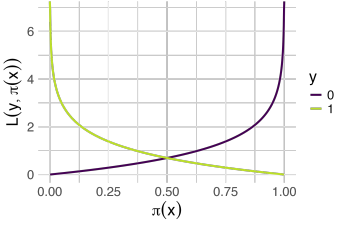
Introduction to Machine Learning

Advanced Risk Minimization
Bernoulli Loss



Learning goals

- Bernoulli (log, logistic, binomial) loss
- Risk minimizer
- Optimal constant
- Complete separation problem

ON PROBS

Likelihood of Bernoulli RV:

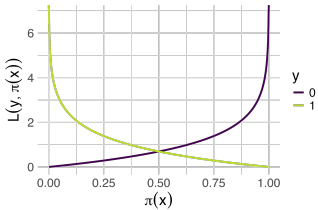
$$\mathcal{L}(\boldsymbol{\theta}) = \prod_{i=1}^{n} \pi\left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}\right)^{y^{(i)}} \left(1 - \pi\left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}\right)\right)^{1-y^{(i)}} \quad , y \in \{0, 1\}$$

Transform into NLL:

$$-\ell(\boldsymbol{\theta}) = \sum_{i=1}^{n} -y^{(i)} \log\left(\pi\left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}\right)\right) - \left(1 - y^{(i)}\right) \log\left(1 - \pi\left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}\right)\right)$$

Bernoulli loss: loss on single sample

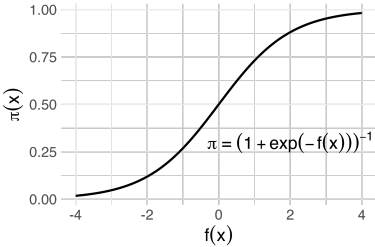
$$L\left(y, \pi(\mathbf{x})\right) = -y \log\left(\pi(\mathbf{x})\right) - (1 - y) \log\left(1 - \pi(\mathbf{x})\right) \quad , y \in \{0, 1\}$$

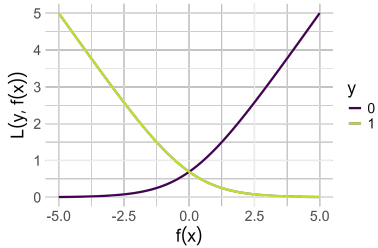


ON DECISION SCORES

- Transform probs into scores (log-odds): $f(\mathbf{x}) = \log\left(\frac{\pi(\mathbf{x})}{1-\pi(\mathbf{x})}\right)$
- Then $\pi(\mathbf{x}) = (1 + \exp(-f(\mathbf{x})))^{-1}$
- Yields equivalent loss formulation

$$L(y, f(\mathbf{x})) = -y \cdot f(\mathbf{x}) + \log(1 + \exp(f(\mathbf{x}))) \quad \text{for } y \in \{0, 1\}$$



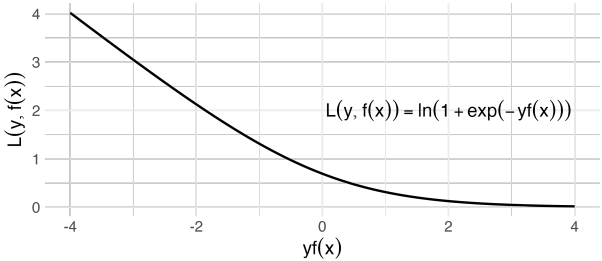


LOSS IN TERMS OF MARGIN

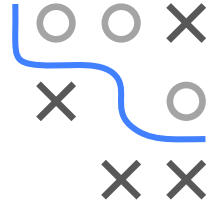
- For $y \in \{-1, +1\}$, loss becomes:

$$L(y, f(\mathbf{x})) = \log(1 + \exp(-y \cdot f(\mathbf{x})))$$

- Simply check both cases for y to see
- Transformation $y' = (y + 1)/2$ and $y' = 2y - 1$ help to convert between the 2 encodings
- All loss variants convex, differentiable



RISK MINIMIZER ON PROBS



For probs and $y \in \{0, 1\}$ is

$$\pi^*(\tilde{\mathbf{x}}) = \eta(\tilde{\mathbf{x}}) = \mathbb{P}(y = 1 \mid \mathbf{x} = \tilde{\mathbf{x}}).$$

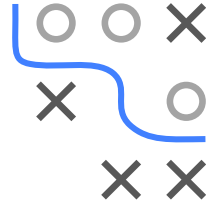
Proof: We had seen before;

$$\mathcal{R}(f) = \mathbb{E}_x \left[L(1, \pi(\mathbf{x})) \cdot \eta(\mathbf{x}) + L(0, \pi(\mathbf{x})) \cdot (1 - \eta(\mathbf{x})) \right]$$

For fixed $\mathbf{x}$, minimize inner part pointwise, use c for best value:

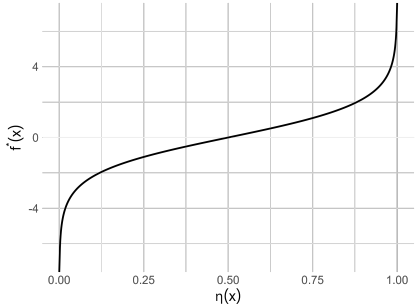
$$\begin{aligned}
\frac{d}{dc} \left(-\log c \cdot \eta(\mathbf{x}) - \log(1 - c) \cdot (1 - \eta(\mathbf{x})) \right) &= 0 \\
-\frac{\eta(\mathbf{x})}{c} + \frac{1 - \eta(\mathbf{x})}{1 - c} &= 0 \\
\frac{-\eta(\mathbf{x}) + \eta(\mathbf{x})c + c - \eta(\mathbf{x})c}{c(1 - c)} &= 0 \\
c &= \eta(\mathbf{x})
\end{aligned}$$

RISK MINIMIZER ON SCORES



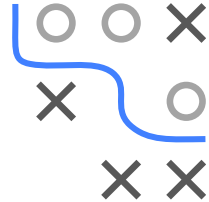
- For $y \in \{-1, 1\}$ and scores $f(\mathbf{x})$: RM is point-wise log-odds

$$f^*(\mathbf{x}) = \log\left(\frac{\eta(\mathbf{x})}{1-\eta(\mathbf{x})}\right)$$



- Undefined for $\eta(\mathbf{x}) \in \{0, 1\}$
- Predicts smooth curve increasing in $\mathbb{P}(y = 1 \mid \mathbf{x} = \tilde{\mathbf{x}})$ otherwise
- Equals 0 when $\mathbb{P}(y = 1 \mid \mathbf{x} = \tilde{\mathbf{x}}) = 0.5$

BERNOULLI: OPTIMAL CONSTANT MODEL



- Optimal constant probability model $\pi(\mathbf{x}) = \theta$ w.r.t. the Bernoulli loss for labels from $\mathcal{Y} = \{0, 1\}$ is

$$\hat{\theta} = \arg\min_{\theta} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^{n} y^{(i)}$$

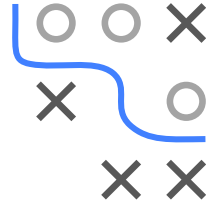
- Fraction of class-1 observations in the observed data
- Derived by setting derivative of risk to 0 and solving for θ
- Optimal constant score model $f(\mathbf{x}) = \theta$ w.r.t. the Bernoulli loss labels from $\mathcal{Y} = \{-1, +1\}$ or $\mathcal{Y} = \{0, 1\}$ is

$$\hat{\theta} = \arg\min_{\theta} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) = \log \frac{n_+}{n_-} = \log \frac{n_+/n}{n_-/n}$$

  where n_- and n_+ are the numbers of negative and positive observations.
- Shows tight (and unsurprising) connection of this loss to log-odds. Proof is left as an exercise.

NAMING CONVENTIONS

We have seen several closely related loss functions:

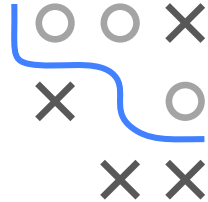


$$
\begin{aligned}
L(y, f(\mathbf{x})) &= \log(1 + \exp(-y f(\mathbf{x}))) && \text{for } y \in \{-1, +1\} \\
L(y, f(\mathbf{x})) &= -y \cdot f(\mathbf{x}) + \log(1 + \exp(f(\mathbf{x}))) && \text{for } y \in \{0, 1\} \\
L(y, \pi(\mathbf{x})) &= -y \log(\pi(\mathbf{x})) - (1 - y) \log(1 - \pi(\mathbf{x})) && \text{for } y \in \{0, 1\} \\
L(y, \pi(\mathbf{x})) &= -\frac{1 + y}{2} \log(\pi(\mathbf{x})) - \frac{1 - y}{2} \log(1 - \pi(\mathbf{x})) && \text{for } y \in \{-1, +1\}
\end{aligned}
$$

They are equally referred to as Bernoulli-, Binomial-, logistic-, log-, or cross-entropy loss.

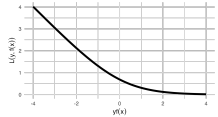
OPTIMIZATION PROPERTIES: CONVERGENCE

In case of **complete separation**, optimization might fail, e.g.:



- **Bernoulli loss** is margin-based and strictly monotonically decreasing in $y \cdot f(\mathbf{x})$:

$$L\left(y, f(\mathbf{x})\right) = \log\left(1 + \exp\left(-yf(\mathbf{x})\right)\right)$$



- f linear in $\boldsymbol{\theta}$, e.g., **logistic regression** with $f(\mathbf{x} \mid \boldsymbol{\theta}) = \boldsymbol{\theta}^\top \mathbf{x}$
- Data perfectly separable by our learner, so we can find $\boldsymbol{\theta}$:

$$y^{(i)} f\left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}\right) = y^{(i)} \boldsymbol{\theta}^T \mathbf{x}^{(i)} > 0 \;\; \forall \mathbf{x}^{(i)}$$

- Can now a construct a strictly better $\boldsymbol{\theta}$

$$\mathcal{R}_{\text{emp}}(2 \cdot \boldsymbol{\theta}) = \sum_{i=1}^{n} L\left(2y^{(i)} \boldsymbol{\theta}^T \mathbf{x}^{(i)}\right) < \mathcal{R}_{\text{emp}}(\boldsymbol{\theta})$$

- As $||\boldsymbol{\theta}||$ increases, sum strictly decreases, as argument of L is strictly larger
- We can iterate that, so there is no local (or global) optimum, and no numerical procedure can converge